1. A coupled partial melting Stokes-Darcy system. Strong form

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla(\phi_f^{-1/2} \tilde{q}_f) &= 0 \\
\mu_s \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2} q) &= 0 \\
\nabla q - \nabla \cdot \hat{\sigma}(\mathbf{v}_s) &= -(1-\phi_f)\rho_r \mathbf{g} \\
\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2} q) &= 0
\end{aligned}
\tag{1.1}
$$

List SI units
q $=$ pascal.
$\mathbf{v}$ $=$ m/s
Non dimensionalization choice

- $\check{\mathbf{v}} = \frac{\tilde{\mathbf{v}_r}}{k_0 \rho_r g_y / \mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = (\frac{k_0 \mu_s}{\mu_f})^{1/2}$

The resulting non dimensionalized form is therefore

$$
\begin{aligned}
\check{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla(\phi_f^{-1/2} \check{q}_f) &= 0 \\
\phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\check{q}_f - \phi_f^{1/2} \check{q}) &= 0 \\
\nabla \check{q} - \nabla \cdot \hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f)\rho_r \hat{\mathbf{g}} \\
\nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\check{q}_f - \phi_f^{1/2} \check{q}) &= 0
\end{aligned}
\tag{1.2}
$$

scaled nondimensionalized weak form let $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla \cdot \mathbf{v}\mathrm{I}$

$$
\begin{aligned}
(2\phi_s \check{\sigma}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) &= -(\phi_s \check{\mathbf{f}}, \Psi_s) \\
(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s} \check{q}_h, \omega\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_{f,h}, \omega\right) &= 0 \\
(\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \Psi_r)) &= 0 \\
(\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \Psi_r), \omega_f) + \left(\frac{1}{\phi_s} \check{q}_{f,h}, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_h, \omega_f\right) &= 0
\end{aligned}
\tag{1.3}
$$

2. Linear system and Uzawa type algorithm. original linear system

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & -\mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & -\mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \tag{2.1}$$

This is a double saddle point system. Symmetrically permutate original linear system and form a enlarged saddle point system.

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & -\mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & -\mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \tag{2.2}$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \tilde{\mathbf{B}} = \begin{bmatrix} \mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_d \end{bmatrix} \quad \tilde{\mathbf{C}} = \begin{bmatrix} \mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & \mathbf{C}_d \end{bmatrix} \tag{2.3}$$

In this newly formed system, the $\tilde{\mathbf{A}}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration.

2.1. Convergence and stability analysis.

3. Test cases. We test the linear system a little bit different from the linear system above.

3.1. Constant porosity and balanced pressure (divergence free velocity field).

3.2. Const porosity and unbalanced pressure.

$$\check{q} = 0 \qquad \check{q}_f = 2(x+y) \qquad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \qquad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \tag{3.1}$$

$$\nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y) \tag{3.2}$$

$$\mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \qquad (\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I} = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix} \tag{3.3}$$

$$\begin{aligned} \tilde{\mathbf{v}}_r + \phi_f^{1/2}\nabla\check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2}\begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2}\nabla\cdot(\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}\check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f}2(x+y) = 0 \\ \nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}\check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\ -2(1-\phi_f)\nabla\cdot\sigma(\check{\mathbf{v}}_s) &= -2(1-\phi_f)\nabla\cdot(\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla\cdot\check{\mathbf{v}}_s)\mathrm{I}) = -\frac{8\phi_f^{1/2}}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned} \tag{3.4}$$